

SOLIDWORKS SYLLABUS		
OBJECTIVE: The objective of this training Session is to impart knowledge for the students about the mechanical part modelling, assembling and presenting in sheets for manufacturing in the industry. The training session also covers a huge area of design and drafting as per the industry norms		
Duration:		
S.NO	CONTENT	HOURS (L/P)
1	Introduction to CAD	
	Features of Solid works	
	Various products available in SolidWorks for Product Design & Assembly Modelling.	
2	SolidWorks Graphical User Interface	
	Feature manager design tree	
	Callouts	
	Handles	
	Confirmation Corner	
	Mouse buttons	
	Keyboard shortcuts	
	Command Manager	
3	Hardware & software requirements	
	Sketch Entities	
	Inference Line	
	Centerline, line	
	Circle, Arc, Ellipse, Rectangle	
	Slots, Polygon, Spline, Points, Text, Construction Geometry	
	Sketch Tools	
	Fillet, Chamfer, Offset, convert Entities	
	Trim, Extend, Mirror, Dynamic Mirror	
4	Move, Copy, Rotate, Scale, Stretch, Sketch pattern	
	Relations	
	Adding Sketch Relation, automatic relations	
	Dimensioning	
	Smart, Horizontal, Vertical, Ordinate, Horizontal ordinate	
	vertical ordinate, align Ordinate, Fully Define Sketch	
	Part Modeling Tools	
5	Creating Reference Planes	
	Creating Extrude Features	
	Direction1, Direction2, From Option, Thin Feature, applying Draft, Selecting Contours	
	Creating Revolve Features	
	Creating Swept Features	
	Creating Loft Features	
	Creating Reference	
	Creating Curves	
6	Creating Fillet Features	
	Inserting Hole Types	
	Creating Chamfer	
	Creating Shell	
	Creating rib	

	Creating Pattern	
7	Environment & Utilities	
	Assembly Modeling Tools	
	Introduction to Assembly Modeling & Approaches	
	Applying Standard Mates	
	Applying Advanced Mates	
	Applying Mechanical Mates	
	Manipulating Components	
	Creating Explode Views	
8	Surface Modeling tools	
	Creating Extrude, Revolve, Swept, Loft, Boundry Surface	
	Inserting Planar Surface, Offset Surface, Radiate Surface	
	Extending a Surface, Surface fill, Ruled Surface, Trimming Surface,	
	Mid Surface, replace Face, Delete Face, Untrim Surface, Knit Surface	
	Thickening a Surface, Move Face	
	Generating Drawing Views	
9	Introduction to Angle of Projection	
	Generating Views	
	Generating Model View	
	Projected Views	
	Inserting Standard 3 View	
	creation relative to model	
	Inserting predefined views	
	Auxiliary Views	
	Detailed Views	
	Crop view	
	Broken –Out Section	
	Broken Views	
	Section View	
10	Alternate Position View	
	Drawing properties	
	Creating Dimensions	
	Smart, Horizontal, Vertical, Baseline, Ordinate	
	Horizontal Ordinate, Vertical Ordinate, Chamfer, Align Collinear	
	Align Parallel/Concentric, Model Dimensions, Auto dimension	
	Annotations, Spell check	
	Inserting Annotations	
11	Datum Features, Hole Callout, Datum Target, Area Hatch, Balloon	
	Centre Mark, Centre Lines, Bill Of Materials	
	Sheets And Templates, Sheet Format.	
	Sheet Metal Design	
	Concepts in Sheet metal design bend allowance bend deduction	
	K-factor Inserting Base Flange, Sheet Metal Tab	
	Edge Flange, Miter Flange, Hem, Jog	
	Creating Break Corner/Corner Trim, Closed Corners, Rip	
	Inserting Sketched Bend, Fold/Unfold, Forming Tools	
	Inserting Cross Break, Welded Corner.	
	Adding Corner Trim, Lofted Trim	
	Conversion Of Solid Body To Sheet Metal.	
	Rendering	